

UE24CS352A Machine Learning - Mini-Project

Predicting Myocardial Infarction Risk using Supervised and Unsupervised Machine Learning

Tharunyah Sudhakar

PES2UG24CS561

Thejaswi S

PES2UG24CS562

Problem Statement:

Myocardial infarction is a leading cause of death, and early risk stratification can direct preventive care to the patients who need it. We build a classifier that labels a patient as high or low MI risk from 13 routine clinical measurements, and we use unsupervised methods like PCA, K-Means, DBSCAN, to test whether risk-related groups emerge without using the labels.

Dataset:

The analysis utilizes the public Kaggle Heart Attack Analysis & Prediction Dataset, comprising 303 patient records with 13 clinical features and a binary target indicating high risk (1). Removing a single duplicate entry leaves 302 patients, of whom approximately 54% are categorized as high risk and 68% are male. The dataset includes 13 distinct features: age, sex, chest pain type (cp), resting blood pressure (trtbps), serum cholesterol (chol), fasting blood sugar (fbs), resting electrocardiographic results (restecg), maximum heart rate achieved (thalachh), exercise-induced angina (exng), ST depression induced by exercise relative to rest (oldpeak), the slope of the peak exercise ST segment (slp), the number of major vessels (caa), and thallium stress test result (thall). While there are no missing values, two anomalies exist: caa includes a value of 4 (outside the standard 0–3 range) and thall includes a value of 0 (outside the standard 1–3 range). Both values were retained, and thall was one-hot encoded to represent 0 as a distinct category..

Approach:

1. Data Preprocessing

Continuous features (age, trtbps, chol, thalachh, and oldpeak) undergo standardization, whereas categorical variables (cp, restecg, slp, and thall) are one-hot encoded. Binary features and caa pass through directly. To prevent test data leakage, all transformation procedures are configured within scikit-learn Pipelines and fitted solely on the training folds.

2. Unsupervised Learning Analysis

Principal Component Analysis (PCA), K-Means clustering ($k = 2$, with PCA components tuned between 2 and 10), and DBSCAN (grid search across eps and min_samples) were applied to the scaled feature set. Target labels were withheld during training and only utilized post-hoc to evaluate cluster separation and alignment via Silhouette score and the Adjusted Rand Index (ARI).

3. Supervised Modeling Framework

Using a stratified 80/20 train-test split (seed 42), the dataset was divided into 242 training instances and 61 testing instances. Nine distinct algorithm families were hyperparameter-tuned via grid search with 5-fold, 2-repeat stratified cross-validation on the training set: logistic regression, elastic net, gradient boosting, bagging, random forest, SVM variants (linear, RBF, polynomial), and a multilayer perceptron. Cross-validated F1-score served as the primary selection metric, with the holdout test set evaluated once for final performance reporting. Cross-validation was selected over a single validation set to mitigate evaluation variance on the 302-sample dataset.

Implementation Overview:

The end-to-end technical pipeline is implemented in Python 3, drawing upon scikit-learn, pandas, seaborn, and matplotlib. To ensure maintainability and reproducibility, the codebase follows a modular structure: standard data ingestion and preprocessing workflows are encapsulated in `common.py`; `eda.py` handles exploratory analysis by producing correlation heatmaps, feature histograms, and boxplots; `clustering.py` runs the unsupervised modules (PCA, K-Means, and DBSCAN); `train.py` drives hyperparameter optimization, model benchmarking, feature importance evaluation, as well as ROC/PR curve and confusion matrix generation; and `predict.py` serves as an inference interface that loads the saved model pipeline to evaluate new patient samples. Executing these scripts sequentially accurately regenerates all figures and empirical results reported in this paper.

Results:

1. Clustering

Scaling matters: K-Means reached ARI 0.30 (3 PCA components; 0.285 with 2) and DBSCAN ARI 0.16 (many points labelled noise), with K-Means silhouette 0.33-0.41. The two K-Means groups are interpretable: one is younger, with higher peak heart rate and few vessel findings (80% labelled 1); the other is older, with more exercise angina, ST depression and vessel findings (25% labelled 1). Agreement is only moderate, which motivates supervised learning.

Figure 1. PCA projection: true outcome, K-Means (clustered on 3 PCs) and DBSCAN (grey = noise).

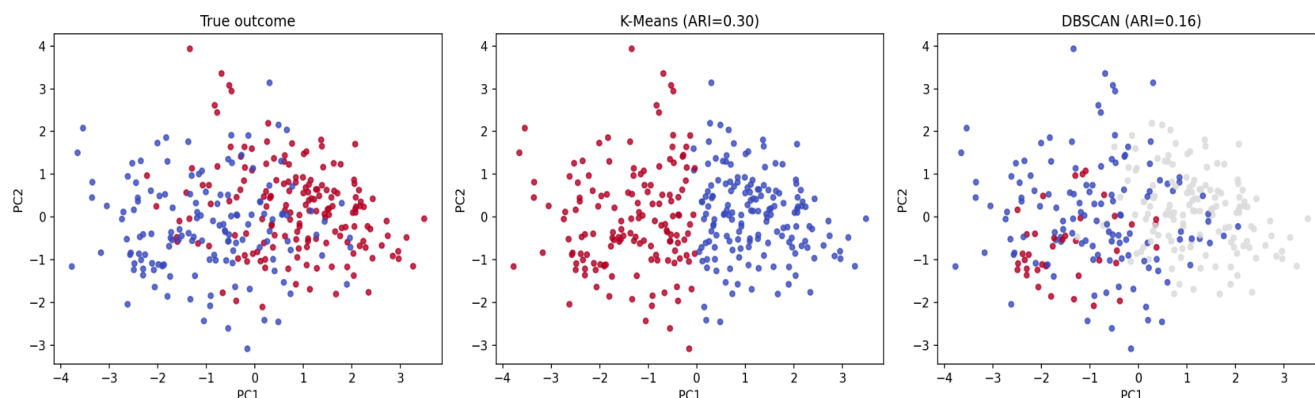


Table 1. Cross-validated F1 (training set, used for selection) and test-set metrics (61 patients).

Model	CV F1	Test AUROC	Recall	Precision	F1	Accuracy
Elastic Net (selected)	0.871	0.885	0.909	0.789	0.845	0.820
Random Forest	0.871	0.890	0.848	0.737	0.789	0.754
SVM (RBF)	0.870	0.893	0.909	0.789	0.845	0.820
Logistic Regression	0.869	0.886	0.909	0.811	0.857	0.836
Gradient Boosting	0.868	0.871	0.879	0.784	0.829	0.803
SVM (linear)	0.866	0.899	0.909	0.789	0.845	0.820
SVM (polynomial)	0.852	0.904	0.939	0.705	0.805	0.754
Bagging	0.833	0.858	0.879	0.784	0.829	0.803
Neural Net (MLP)	0.804	0.812	0.818	0.771	0.794	0.770

2. Supervised models

Elastic Net ($C = 0.1$, L1 ratio = 0.5) was selected as the primary model. On the test set, it achieved an AUROC of 0.885, a recall of 0.909, a precision of 0.789, and an overall accuracy of 0.820 (30 true positives, 20 true negatives, 8 false positives, and 3 false negatives), outperforming the majority-class baseline of 0.540. Prioritizing high recall minimizes missed high-risk cases in clinical screening settings. Permutation importance identified *caa*, *thall*, *thalachh*, *oldpeak*, and *cp* as the most influential features, whereas *chol* contributed minimal predictive value.

Figure 2. Elastic Net: test confusion matrix and permutation importance (AUROC drop).

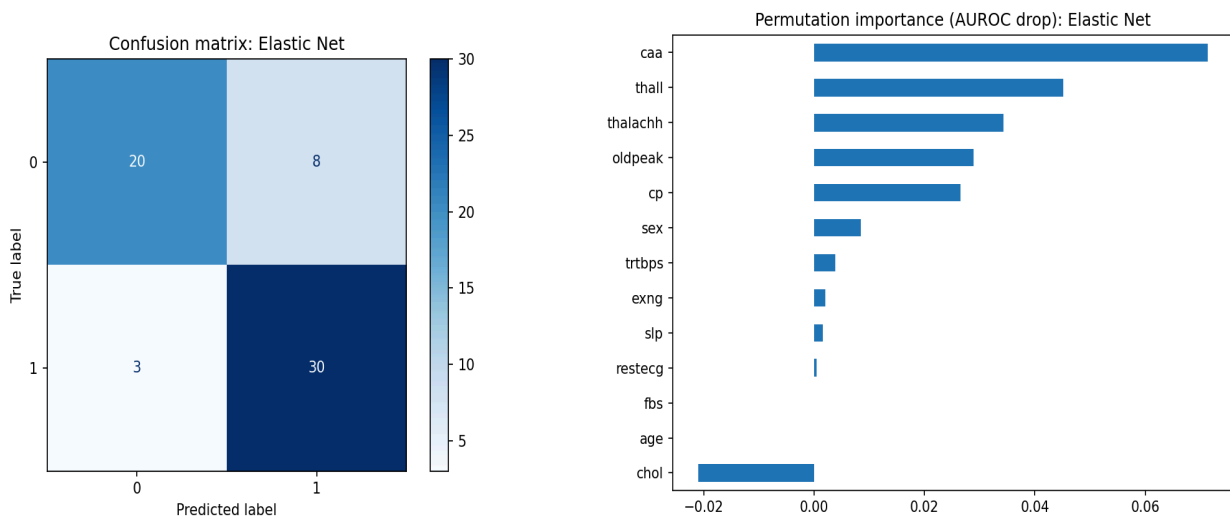
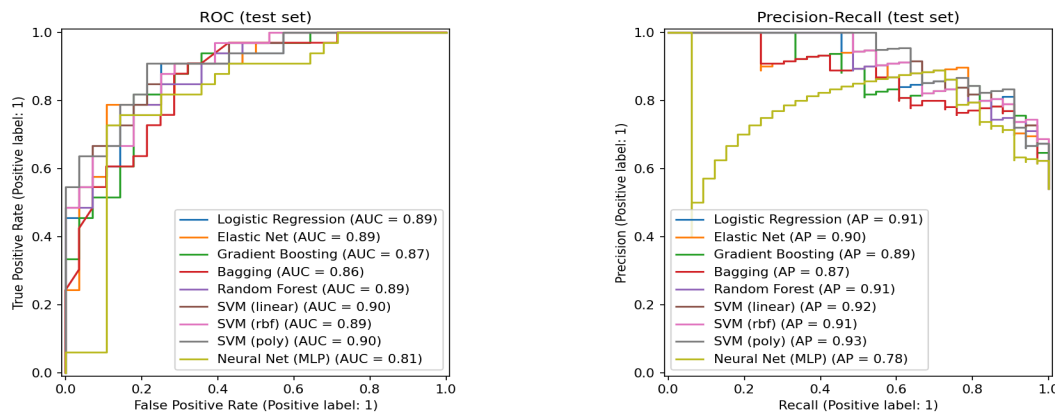


Figure 3. ROC and precision-recall curves for all nine tuned models on the test set; the curves overlap heavily (AUROC 0.81-0.90)



Conclusions and Limitations:

On this small tabular dataset, lightweight regularized linear models match the performance of more complex architectures. The top four models share nearly identical cross-validated F1 scores ranging from 0.868 to 0.871 (standard deviation 0.03–0.05). Conversely, the Multilayer Perceptron (MLP) yields the lowest cross-validated F1 score at 0.80, which aligns with expectations given the limited training size of roughly 240 samples. Elastic Net was selected according to the pre-established selection rule and offers superior interpretability.

Key Limitations:

- **Small Test Sample Size:** With a test set of only 61 patients, a single misclassification shifts overall accuracy by 1.6 percentage points and recall by 3 percentage points. Consequently, performance discrepancies among the top algorithms fall within statistical noise, and no single model can be definitively labeled superior.
- **Potential Label Inversion:** In this dataset version, key clinical risk factors—including exercise-induced angina, ST depression, major vessel count, and male sex—correlate with lower output values. This contradicts established medical domain knowledge and points to possible label inversion. The original target coding was preserved because it does not alter predictive performance metrics, though it directly impacts clinical interpretability.
- **Scope and Generalizability:** The records originate from a single small cohort and lack external validation. As such, this implementation serves strictly as an educational study and is not suitable for clinical decision-making.

Future Work: Subsequent enhancements will incorporate bootstrap confidence interval estimation, nested cross-validation pipelines, evaluation on external datasets, and probability calibration.